

Ex. No - 06

1. C Programs to display a message TNAU motto "Till, Toil, Triumph".

```
#include<stdio.h>
#include<conio.h>
void main( )
{
printf("Till, Toil, Triumph");
getch( );
}
```

Output:

Till,Toil,Triumph

2. C program for Leaf Area Index

```
#include<stdio.h>
#include<conio.h>
void main( )
{
int la, ga;
float lai;
clrscr( );
printf("Enter Leaf Area\n");
scanf("%d", &la);
printf("Enter Ground Area\n");
scanf("%d", &ga);
lai = la / ga;
printf("Leaf Area = %d\n", la);
printf("Ground Area = %d\n", ga);
printf("lai = %f\n", lai);
getch( );
}
```

Output:

Enter Leaf area
56
Enter Ground Area
32
Leaf Area = 56
Ground Area = 32
Lai = 1.000000

3. C Program for finding CGR

```
#include<stdio.h>
#include<conio.h>
void main( )
{
int w1, w2, t1,t2, plantweight, plantday;
float cgr;
clrscr( );
printf("Enter Plant Weights\n");
scanf("%d%d", &w2, &w1);
printf("Enter Plant Count (Days)\n");
scanf("%d%d", &t2, &t1);
plantweight = w2 - w1;
plantday = t2 - t1;
cgr = plantweight / plantday;
printf("Plant Weight = %d\n", plantweight);
printf("Plant Day = %d\n", plantday);
printf("Crop Growth Rate = %f\n", cgr);
getch( );
}
```

Output:

```
Enter Plant Weights
34 12
Enter Plant Count (Days)
34 12
Plant Weight = 22
Plant Day = 22
Crop Growth Rate = 1.000000
```

4. C program for finding the greatest average seed sales of two districts during samba season.

```
#include<stdio.h>
#include<conio.h>
void main( )
{
int avg1, avg2;
printf("Enter two districts average values:\n");
scanf("%d %d",&avg1, &avg2);
if(avg1 > avg2)
printf("%d District one is greatest seed sale",avg1);
else if(avg2 > avg1)
printf("%d District two is greatest seed sale",avg2);
else
```

```
printf("%d and %d District one & two are equal", avg1, avg2);  
getch( );  
}
```

Output:

Enter two districts average values:

110

140

140 District two is greatest seed sale.

5. C program to convert the temperature from Fahrenheit to centigrade.

```
#include<stdio.h>  
#include<conio.h>  
void main( )  
{  
float celsius,fahrenheit;  
printf("\nEnter temperature in Fahrenheit:\n");  
scanf("%f", &fahrenheit);  
celsius = fahrenheit - 32) * 5/9;  
printf("\nCelsius = %.3f", celsius);  
getch( );  
}
```

Output:

Enter temperature in Fahrenheit:

120

Celsius = 48.889

6. C program to find the Fibonacci series till term ≤ 50 .

```
#include <stdio.h>  
#include<conio.h>  
void main( )  
{  
int i, n, t1 = 0, t2 = 1, nextTerm;  
printf("Enter the number of terms: ");  
scanf("%d", &n);  
printf("Fibonacci Series: ");  
for (i = 1; i <= n; ++i)  
{  
printf("%d \t ", t1);  
nextTerm = t1 + t2;  
t1 = t2;  
t2 = nextTerm;  
}  
getch( );  
}
```

Output:

Enter the number of terms: 15
Fibonacci Series:0 1 1 2 3 5 8 13 21
34 55 89 144 233 377

7. C program for finding Factorial Number.

```
#include <stdio.h>
#include<conio.h>
void main( )
{
    int n, i;
    unsigned long long fact = 1;
    printf("Enter an integer: ");
    scanf("%d", &n);
    // shows error if the user enters a negative integer
    if (n < 0)
        printf("Error! Factorial of a negative number doesn't exist.");
    else
    {
        for (i = 1; i <= n; ++i)
        {
            fact *= i;
        }
        printf("Factorial of %d = %llu", n, fact);
    }
    getch( );
}
```

Output:

Enter an integer : 5
Factorial of 5 = 120

1. Find total, maximum, minimum and average rainfall of last five years in our District.

```
#include<stdio.h>
#include<conio.h>
void main( )
{
    int i, n, sum = 0, rainfall, max = 0, min;
    float average;
    clrscr( );
    printf("\n Enter the number of years: \n");
    scanf("%d", &n);
```

```

printf("\n Enter the number of rainfall in the given years: \n");
for(i=0; i<n; i++)
{
    for(i=0; i<n; i++)
    {
        scanf("%d", &rainfall);
        sum = sum + rainfall;
        if(rainfall < min)
        {
            min = rainfall;
        }
        if(rainfall > max)
        {
            max = rainfall;
        }
    }
}
average = sum / n;
printf("\n Sum of the %d year Rainfall = %d", n, sum);
printf("\n Average of the %d year Rainfall = %.2f", n, average);
printf("\n The minimum Rainfall is = %d", min);
printf("\n The maximum Rainfall is = %d", max);
getch( );
}

```

Output:

```

Enter the number of years : 5
Enter the number of rainfall in the given years:
12
13
14
15
16
Sum of the 5 year Rainfall = 70
Average of the 5 year Rainfall = 14.00
The minimum Rainfall is = 12
The maximum Rainfall is = 16

```

2. Write a C program to accept weekly temperature and find the average temperature for Madurai Dist.

```
#include<stdio.h>
#include<conio.h>
void main( )
{
float temparray[7], sum = 0, average;
int i;
printf("\n The Madurai Dist temperature: ");
clrscr( );
for(i=0; i<7; i++)
{
printf("Enter temperature reading of day: %d = ", i+1);
scanf("%f", &temparray[i]);
sum = sum + temparray[i];
}
average = sum / 7.0;
printf("\n The total temperature for weekly =%.2f", sum);
printf("\n The average temperature of the weekly is =%.2f", average);
getch( );
}
```

Output:

```
Enter temperature reading of day:1 = 100
Enter temperature reading of day:2 = 101
Enter temperature reading of day:3 = 102
Enter temperature reading of day:4 = 103
Enter temperature reading of day:5 = 104
Enter temperature reading of day:6 = 105
Enter temperature reading of day:7 = 106
```

The total temperature for weekly = 721.00

The average temperature of the weekly is = 103.00

3. The Last 10 years rice yield of Sivagangai District using for and while loop.

```
#include<stdio.h>
#include<conio.h>
void main( )
{
int i = 1, n, riceyield, sum = 0;
float average;
clrscr( );
```

```

printf(" \n Enter the number of years: \n" );
scanf("%d", &n);
printf(" \n Enter the last 10 years Rice Yield of Sivagangai Dist: \n");
while(i <= n)
{
printf("%d", i);
i++;
scanf("%d", &riceyield);
sum = sum + riceyield;
}
/* for(i=0; i<n; i++)
{
for(i=0; i<n; i++)
{
scanf("%d", &riceyield);
sum = sum + riceyield;
}
} */
average = sum / n;
printf("\n sum of the %d year Rice yield = %d", n, sum);
printf("\n Average of the %d year Rice yield = %.2f", n, average);
getch( );
}

```

Output:

Enter the number of years:10

Enter the last 10 years Rice Yield of Sivagangai Dist:

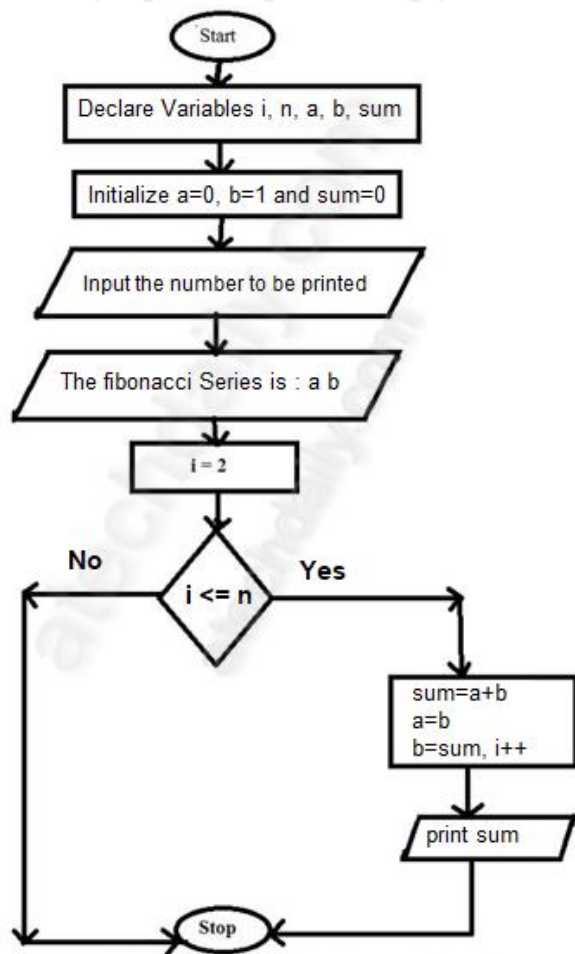
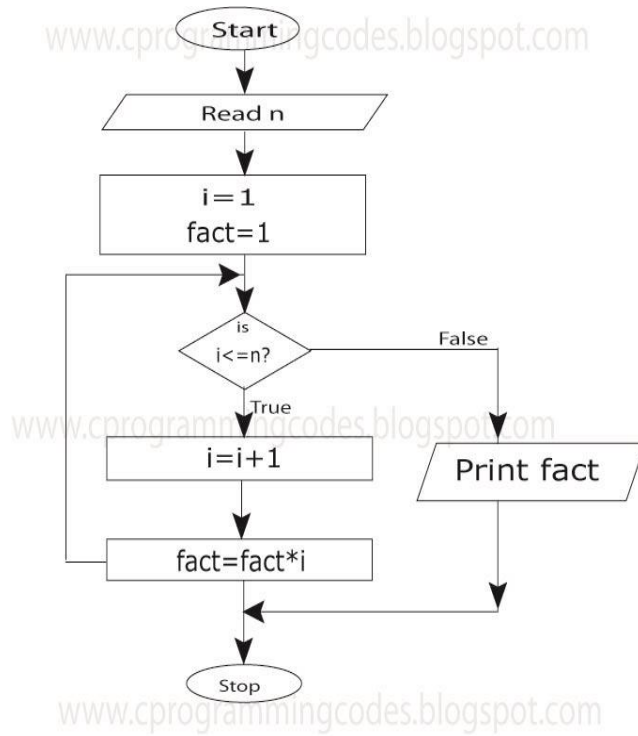
```

1 100
2 120
3 130
4 150
5 160
6 170
7 180
8 190
9 200
10 201

```

Sum of the 10 years Rice yield = 1601

Average of the 10 year Rice yield = 160.1



Flowchart

